

Marine microbiology

Lectures 15-16

Some important types of marine bacteria

References:

Munn. Marine Microbiology (MM): Chapter 4

Prescott's Microbiology (PM): Chapter 21, 22, 23, 24 (selected organisms)

Introduction

Outcomes:

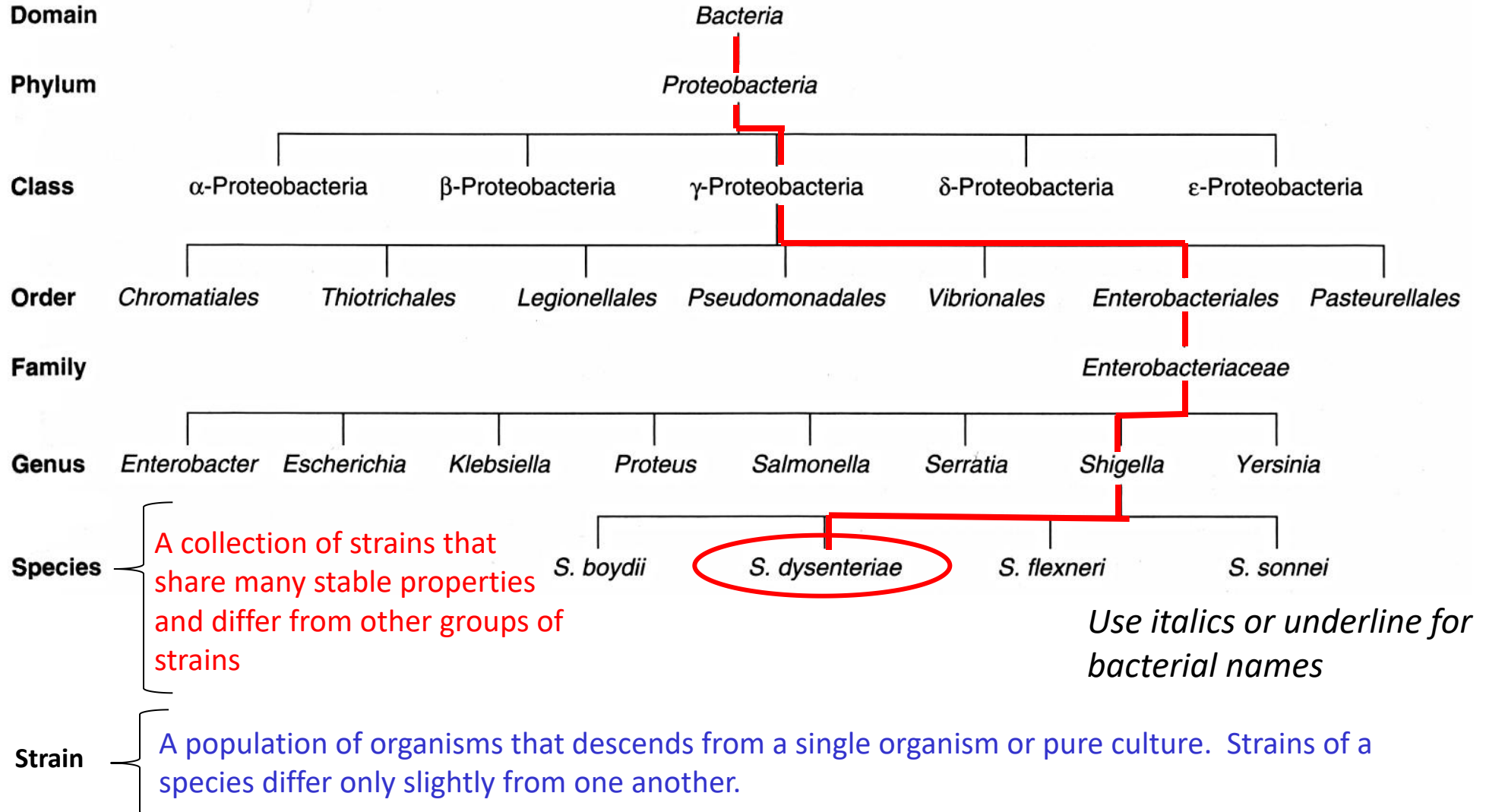
At the end of this lecture you will have a broad understanding of the main types of marine bacteria, and be able to describe:

1. How Gram-negative bacteria are classified into Proteobacteria (变形菌门) and Non-Proteobacteria
2. How Gram-positive bacteria are classified into low G + C and high G + C
3. The main features of some **Gram negative bacteria**
 - Non-Proteobacteria Photosynthetic bacteria
 - Proteobacteria in alpha, gamma and delta groups
 - Magnetotactic (趋磁性) bacteria
 - Purple photosynthetic bacteria
4. The main features of some **Gram positive bacteria**
 - Low G + C members of the *Firmicutes* (厚壁菌门)
 - High G + C members of the *Actinobacteria* (放线菌)

Taxonomic Ranks (a reminder)

Groups at any rank level share common properties with the group they belong to in the rank above

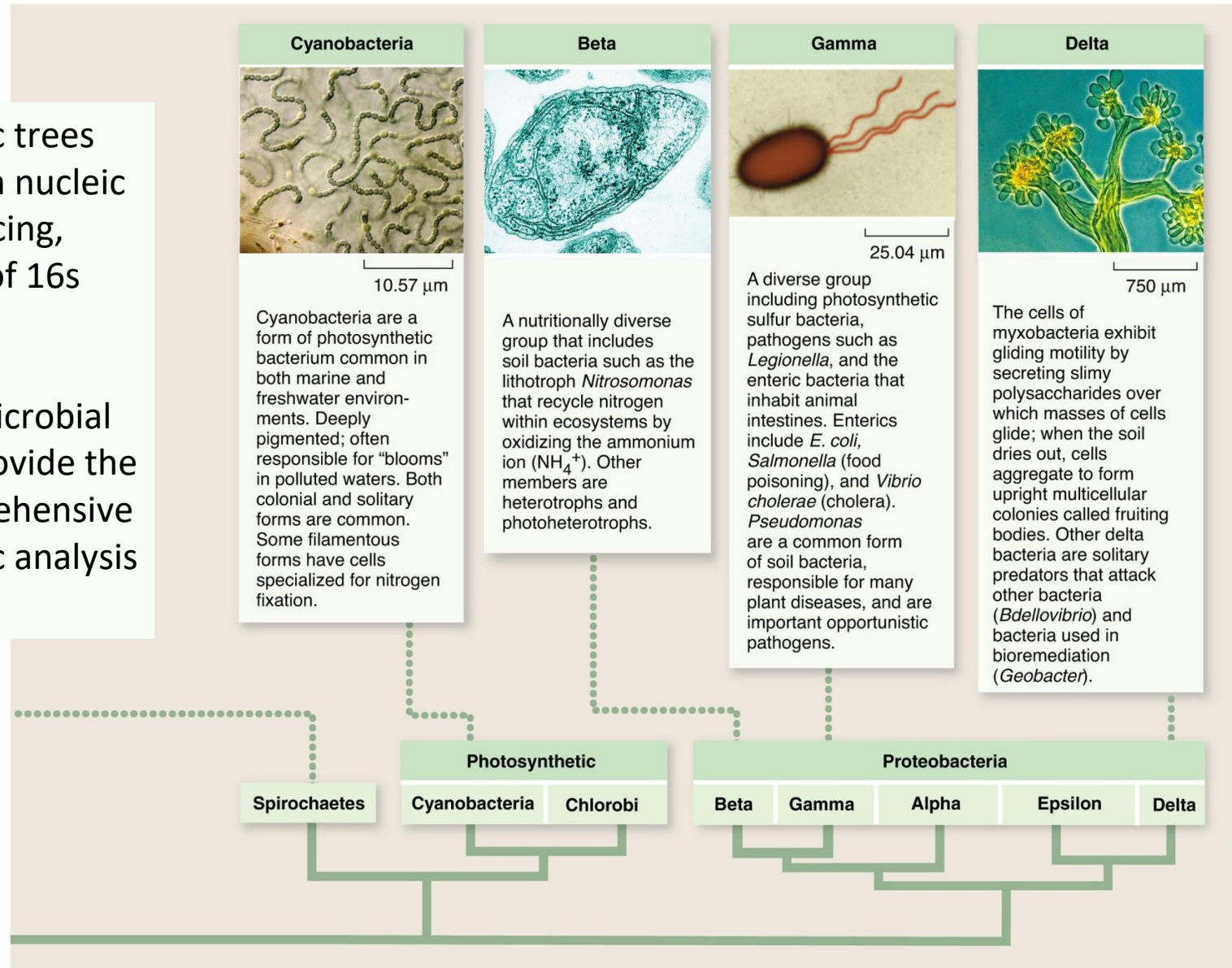
Rank:



Some major groups of Bacteria and Archaea (Prescott Fig 19.14)

Phylogenetic trees are based on nucleic acid sequencing, most often of 16s rRNA

Complete microbial genomes provide the most comprehensive phylogenetic analysis



Diversity of Gram-Negative Bacteria

Two groups based on 16s rRNA studies:

A. Non-Proteobacteria

- Diverse phyla (門)
- Contains photosynthetic cyanobacteria and green sulfur bacteria, spirochaetes (螺旋体属) etc.

B. Proteobacteria

- Named for the Greek sea-god Proteus, who could change his shape - i.e. the largest and most diverse group of bacteria
- Comprises 5 subgroups termed alpha (α), beta (β), gamma (γ), delta (δ) and epsilon (ϵ) proteobacteria
- Sometimes called 'the purple bacteria'

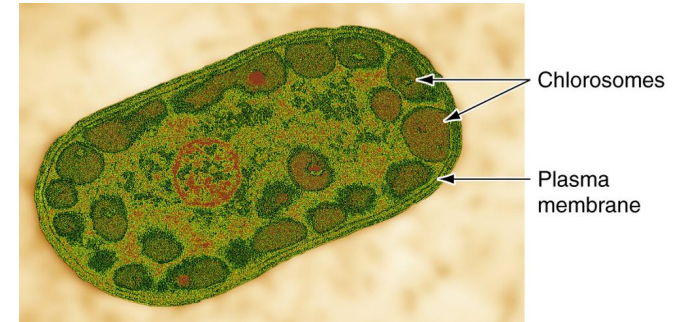


A. Non-Proteobacteria

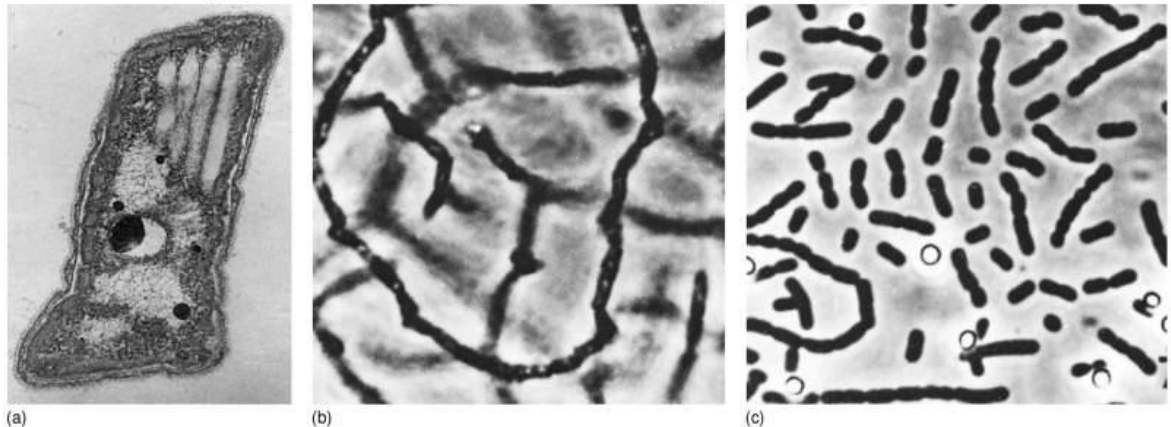
1. Photosynthetic bacteria

(a) Green sulphur bacteria

- anoxygenic (不产氧的) photosynthesis (do not evolve O_2)
- Bacteriochlorophylls are in a membrane-bound structure called the chlorosome
- deposit (沉积) S-granules outside the cell
- Common in microbial mats (微生物席) and sediments with high levels of sulphide



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Green Sulfur Bacteria. (a) chlorosomes in dark grey, gas vesicles in light grey with pointed ends, (b) net-like structure, (c) extracellular S granules.

A. Non-Proteobacteria

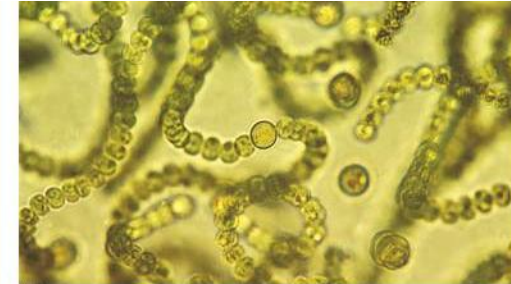
1. Photosynthetic bacteria

(b) Phylum *Cyanobacteria*

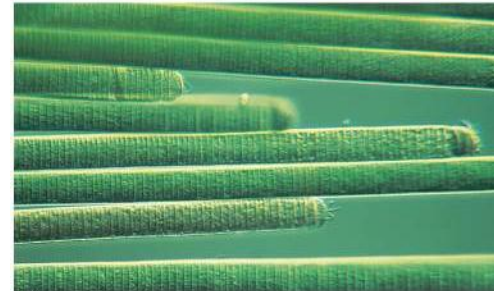
- Largest, most diverse group of photosynthetic bacteria
- Oxygenic photosynthesis (evolve O_2)
- Changed earth's early atmosphere: stimulated aerobic life
- Vary greatly in shape and appearance
 - ◆ unicellular small rods
 - ◆ spirals, chains or groups of cocci
 - ◆ large branching filaments
 - ◆ some form motile filaments that can glide along surfaces, using polysaccharide slime: mechanism not well-understood



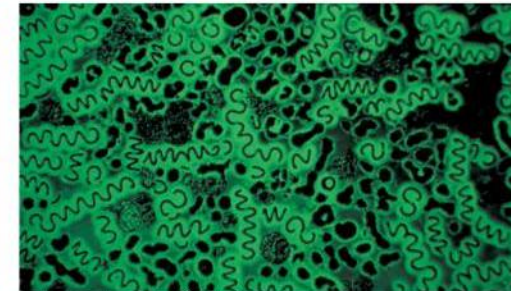
(a) *Chroococcus turgidus*



(b) *Nostoc*



(c) *Oscillatoria*



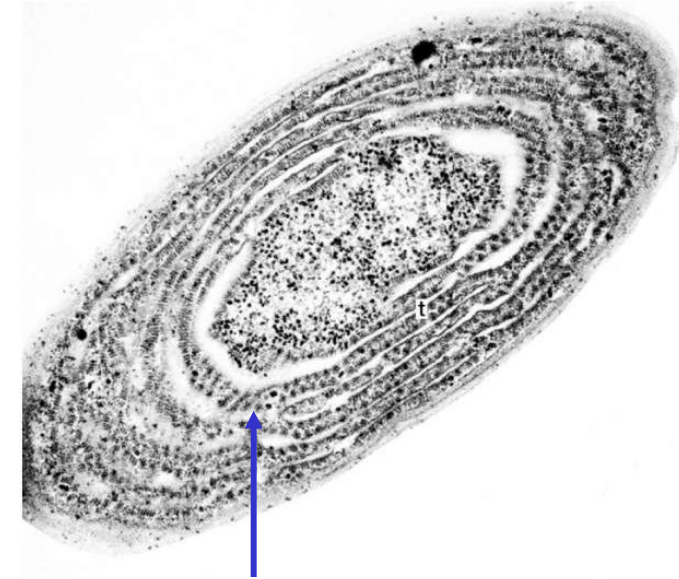
(d) *Anabaena spiroides* and *Microcystis aeruginosa*

A. Non-Proteobacteria

1. Photosynthetic bacteria

(b) Phylum *Cyanobacteria*

- Chlorophylla is contained within **thylakoid** (类囊体) membranes which are lined with **phycobilisomes** (藻胆体): this is where photosynthesis occurs
- Appear blue-green due to presence of blue pigment and chlorophyll
 - ◆ Sometimes referred to as “blue-green algae”- this is **INCORRECT** as they are bacteria
- Genera *Prochlorococcus* (原绿球藻) and *Synechococcus* (聚球藻) dominate the picoplankton of many oceans
 - ◆ *Prochlorococcus*: a very small cyanobacterium 0.6 μm diameter
 - ◆ Most abundant photosynthetic organism on Earth



Thylakoid membranes
(membranes internal to the cell)
lined with phycobilisomes
(black particles on the
membranes containing the
photosynthetic pigments)

A. Non-Proteobacteria

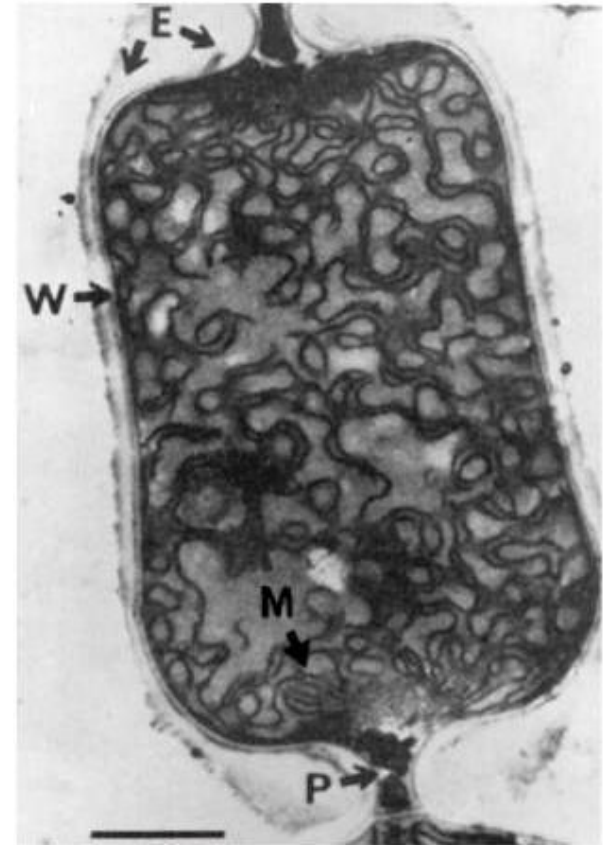
1. Photosynthetic bacteria

(b) Phylum *Cyanobacteria*

- Many marine Cyanobacteria have heterocysts (异形细胞): specialised cells - fix N_2 and convert it to NH_3 to build amino acids



Nostoc 念珠藻 (a major marine genus) with heterocysts



(c) *Anabaena* heterocyst

Anabaena (项圈藻) heterocyst:
electron micrograph

W = cell wall, P = pore channel to
adjacent cell, E = membranes

A. Non-Proteobacteria

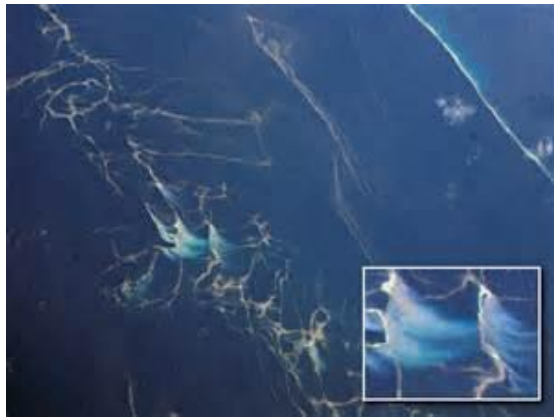
1. Photosynthetic bacteria

(b) Phylum *Cyanobacteria*

- Present in most waters, soils. In marine environment, habitats include plankton, sea ice, shallow sediments, microbial mats
- Can form 'blooms' - fertilizer allows rapid growth
 - may produce toxins that kill livestock and other animals that drink the water

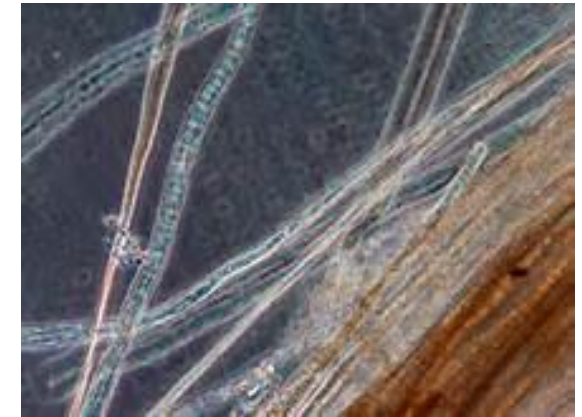


Bloom of Cyanobacteria and Algae in nutrient-rich pond



Trichodesmium 束毛藻 bloom of NW Australia seen from space station

Trichodesmium (a major marine nitrogen-fixer) grows as colonies of 50-200 filaments



A. Non-Proteobacteria

1. Photosynthetic bacteria

(b) Phylum *Cyanobacteria*

- Important in formation of microbial mats in shallow (浅的) water
- **Microbial mats:** communities of various bacteria and protists that form layered structures, can be several mm thick
 - Composition varies, can include diatoms, anoxygenic phototrophs, chemoheterotrophs, as well as *Cyanobacteria*
- Stromatolites (叠层石) are fossilised mats of filamentous (丝状细菌) bacteria and trapped sediment – 3 billion years old



stromatalites

B. Proteobacteria

- Largest, most diverse group of bacteria
 - > 500 genera, 1300 species
- 16s rRNA sequencing revealed 5 subgroups (α , β , γ , δ , ϵ)
- Sometimes called the **purple bacteria** because purple photosynthetic bacteria are found in α , β , γ subgroups



(a)



(b)

Blooms of purple
photosynthetic
bacteria

Alpha (α) proteobacteria

Includes:

- most oligotrophic (low nutrient environment) proteobacteria, dominant (占优势) in the oceans
- some with distinctive morphology (形态学) e.g. *Caulobacter* (柄杆菌)
- some photosynthetic bacteria (purple nonsulphur bacteria)
- SAR 11 clade 进化分支 (cluster): defined by 16S rRNA sequencing, most abundant group of microorganisms in the sea – see later lecture on metagenomics
- *Roseobacter* (玫瑰杆菌) clade: defined by 16S rRNA sequencing, very abundant in plankton

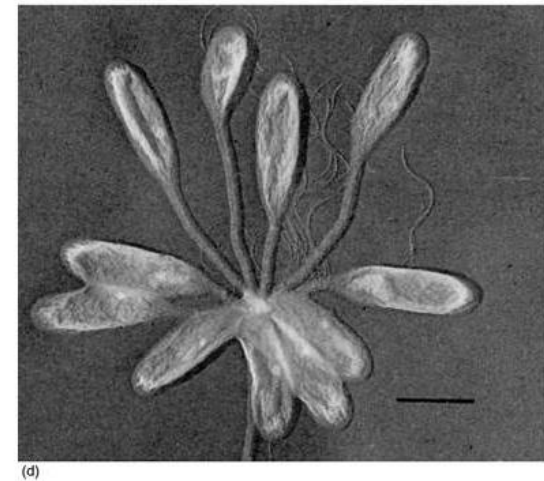
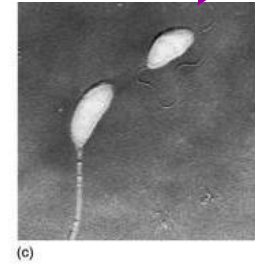
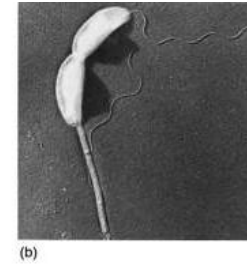
SAR11演化枝的细菌(The SAR11 clade, Pelagibacteraceae)是迄今世界海洋中最丰富的微生物, 因而也是碳循环中一个重要因素。最初之所以这样命名, 是因为它们是在“马尾藻海 Sargasso Sea”中发现的。

Alpha (α) proteobacteria

1. Genus: *Caulobacter*

- Gram-negative, unusual life cycle
 - ◆ unequal cell division, “mother cell” buds off smaller “daughter cell”
- Found in marine and fresh water, soil
- Possess a **prostheca** or **stalk** (柄或茎)
 - ◆ functions in attachment to surfaces (algae, stones etc.)
 - ◆ helps these aerobes to remain in O₂-rich regions
 - ◆ also functions in nutrient uptake

Caulobacter cell divides to produce smaller daughter cell which is motile and swims away



rosettes of *Caulobacter* cells adhering to each other by their prosthecae

Gamma (γ) proteobacteria

- Largest subgroup of proteobacteria
- Extraordinary physiological diversity
- Many can be cultured; were regarded as dominant in oceans, now known not to be so
- Major groupings include:
 - some **photosynthetic** bacteria (**purple sulphur bacteria**)
 - facultatively (兼性) anaerobic chemorganotrophs (e.g. families *Enterobacteriaceae* 肠杆菌, *Vibrionaceae* 弧菌)
 - aerobic chemoorganotrophs (e.g. genus *Pseudomonas* 假单胞菌)
 - chemolithotroph *Thiomargarita namibiensis* (纳米比亚嗜硫珠菌)

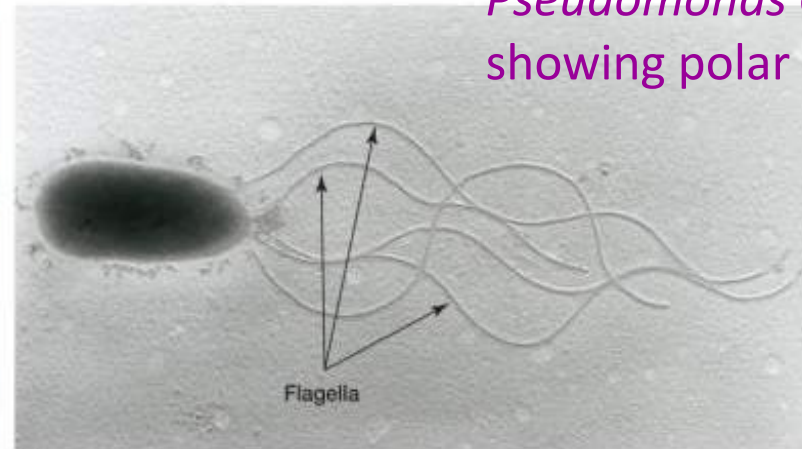
Gamma (γ) proteobacteria

1. Genus: *Pseudomonas*

- Gram-negative rods with aerobic respiration - motile with one or more polar flagella – found in soil and water, coastal marine waters
- Many degrade a wide range of organic substrates
 - ♦ use > 80 different substances as C and energy sources e.g. oils, pesticides (杀虫剂)
- *Ps. aeruginosa* (铜绿假单胞菌) – an important human pathogen, infects people with lowered resistance, burns, wounds



(a)



(b)

Pseudomonas cells
showing polar flagella

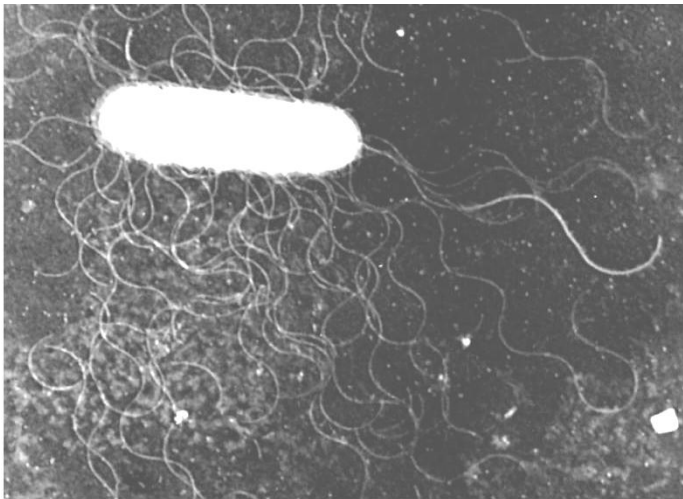
Gamma (γ) proteobacteria

2. Genus: *Vibrio* (弧菌) and related genera *Aliivibrio*, *Photobacterium*

- Gram-negative : curved rods (comma-shaped,,): facultative anaerobes
- Aquatic, particularly marine habitats

e.g. *V. cholerae* (霍乱弧菌) is a major human pathogen:

Cholera – epidemics (流行病), pandemics (outbreaks across different countries)
- watery profuse diarrhoea (腹泻), high mortality rate if untreated



Vibrio with polar and lateral (peritrichous) flagella

Gamma (γ) proteobacteria

2. Genus: *Vibrio* and related genera *Aliivibrio*, *Photobacterium*

- Commonly associated with surfaces of many marine animals and plants, and organic material
- Play major roles in:
 - ◆ Initial colonization and **biofilm** formation on surfaces
 - ◆ **Symbionts** in light organism of marine animals (see later lecture)
 - ◆ **Pathogens** of coral, fish, molluscs (软体动物), crustaceans (甲壳类动物) (see later lecture)
- Some important species:
 - V. alginolyticus* (溶藻弧菌)
 - V. harveyi* (哈维氏弧菌)
 - V. vulnificus* (创伤弧菌)
 - Aliivibrio (Vibrio) fischeri* (费氏弧菌)
 - Photobacterium damsela* (美人鱼发光杆菌)



Squid (鱿鱼)
bioluminescence (生物发光) due to vibrios



Coral white area due to
vibrio infection

Gamma (γ) proteobacteria

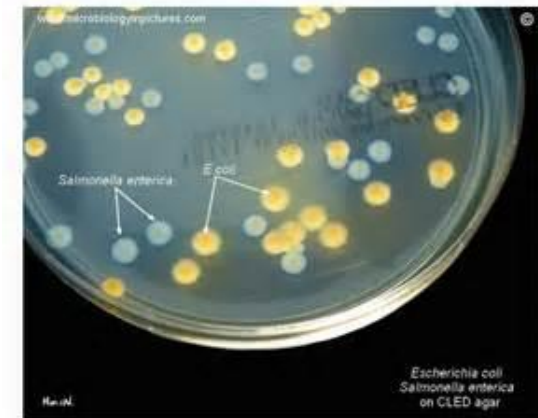
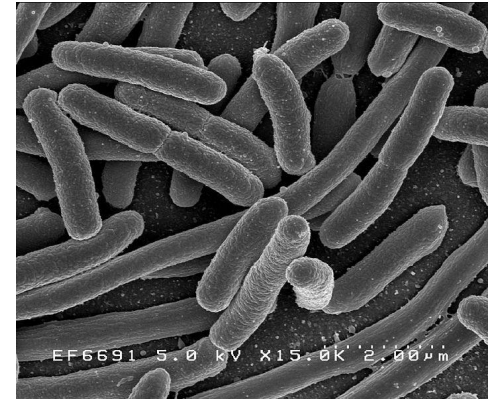
3. Family: *Enterobacteriaceae* (“enterics” or enterobacteria”)

- Gram-negative, straight rods
- Facultative anaerobes
- Simple nutritional requirements - ferment glucose and other carbohydrates
- Mostly motile with peritrichous flagella (or non-motile)
- Inhabit intestinal tracts humans, animals, fish, marine mammals
- Can be isolated from coastal waters containing sewage (污水)
- Important species:

Escherichia (*E. coli*)

Salmonella (沙门氏菌)

Serratia (沙雷氏菌)

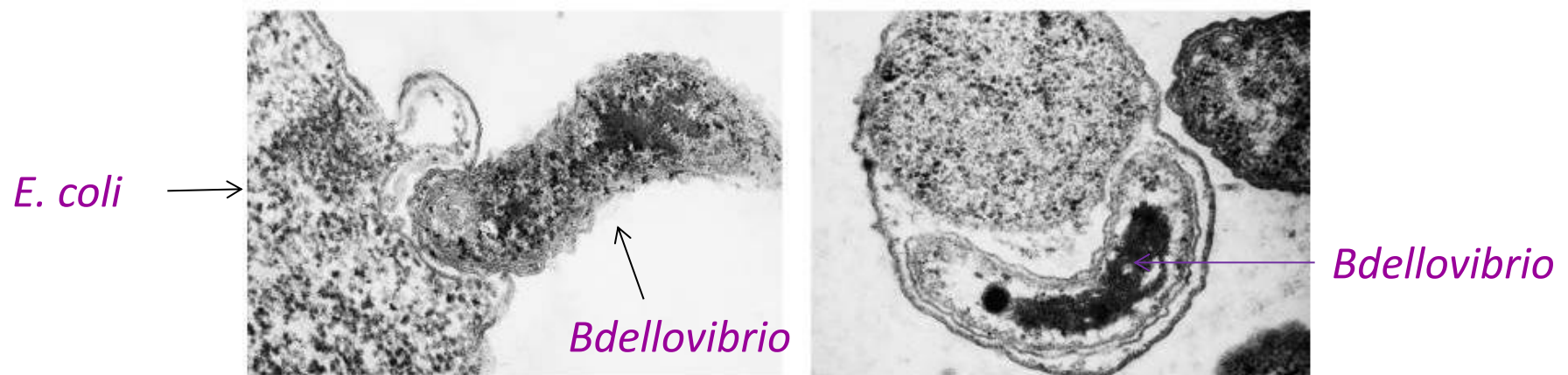


Delta (δ) proteobacteria

Includes some bacterial predators

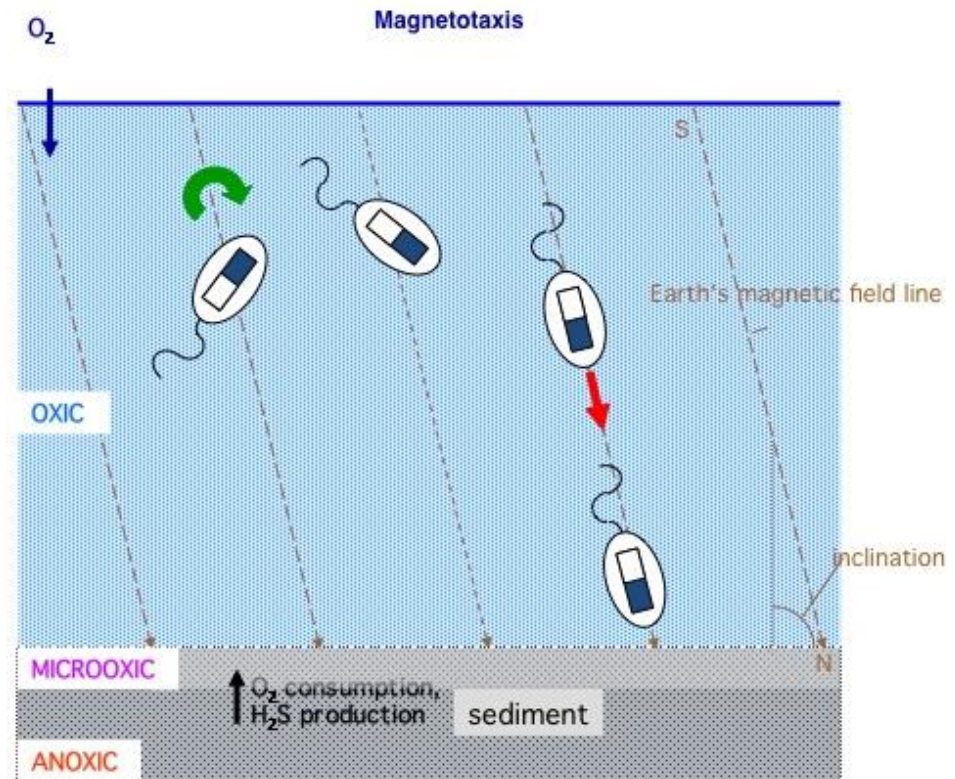
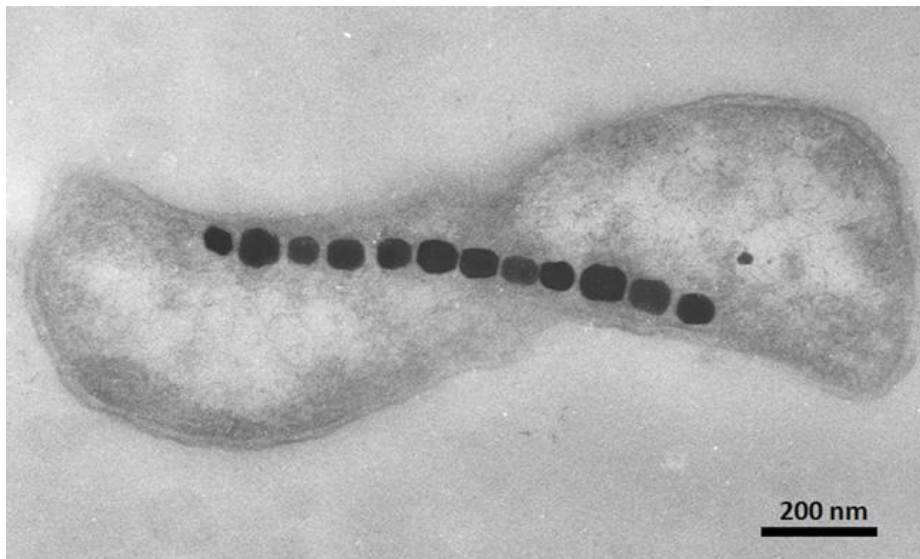
Genus: *Bdellovibrio* 蛭弧菌 (greek: *Bdella* = leech)

- Aerobic - motile Gram-negative – small spiral (螺旋形) cells
- Preys (捕食) on other Gram-negative bacteria
- Collides (碰撞) violently (猛烈地) with prey, rotates at 100 rev/sec, bores hole (钻孔) through cell wall with aid of enzymes
- Grows under cell wall, progeny (子代) released on cell lysis, swim away to seek new hosts
- Widespread in marine environment



Magnetotactic (趋磁性) bacteria occur in various groups e.g. α and δ Proteobacteria

- Cells are rod-shaped or spiral
- Contain chains of magnetic particles made e.g. of magnetite Fe_3O_4
- Cell aligns with Earth's magnetic field, and bacteria swim in this direction using polar flagella
- Bacteria are usually microaerophilic (微需氧的), seeking suitable zone in sediments with optimal O_2 and sulfide



Purple bacteria occur in various groups e.g. α , β and δ Proteobacteria

■ Several groups of bacteria carry out anoxygenic photosynthesis

A. Purple sulfur bacteria

- Use H_2S as electron source
- S granules are deposited in the cells
- Anaerobes, found in shallow marine sediments



B. Purple nonsulfur bacteria

- Grow anaerobically as photoorganoheterotrophs, using organic compounds for e- and C
- Grow aerobically in absence of light

C. *Roseobacter* clade (cluster)



Green and purple bacteria

Classification of Gram-Positive Bacteria

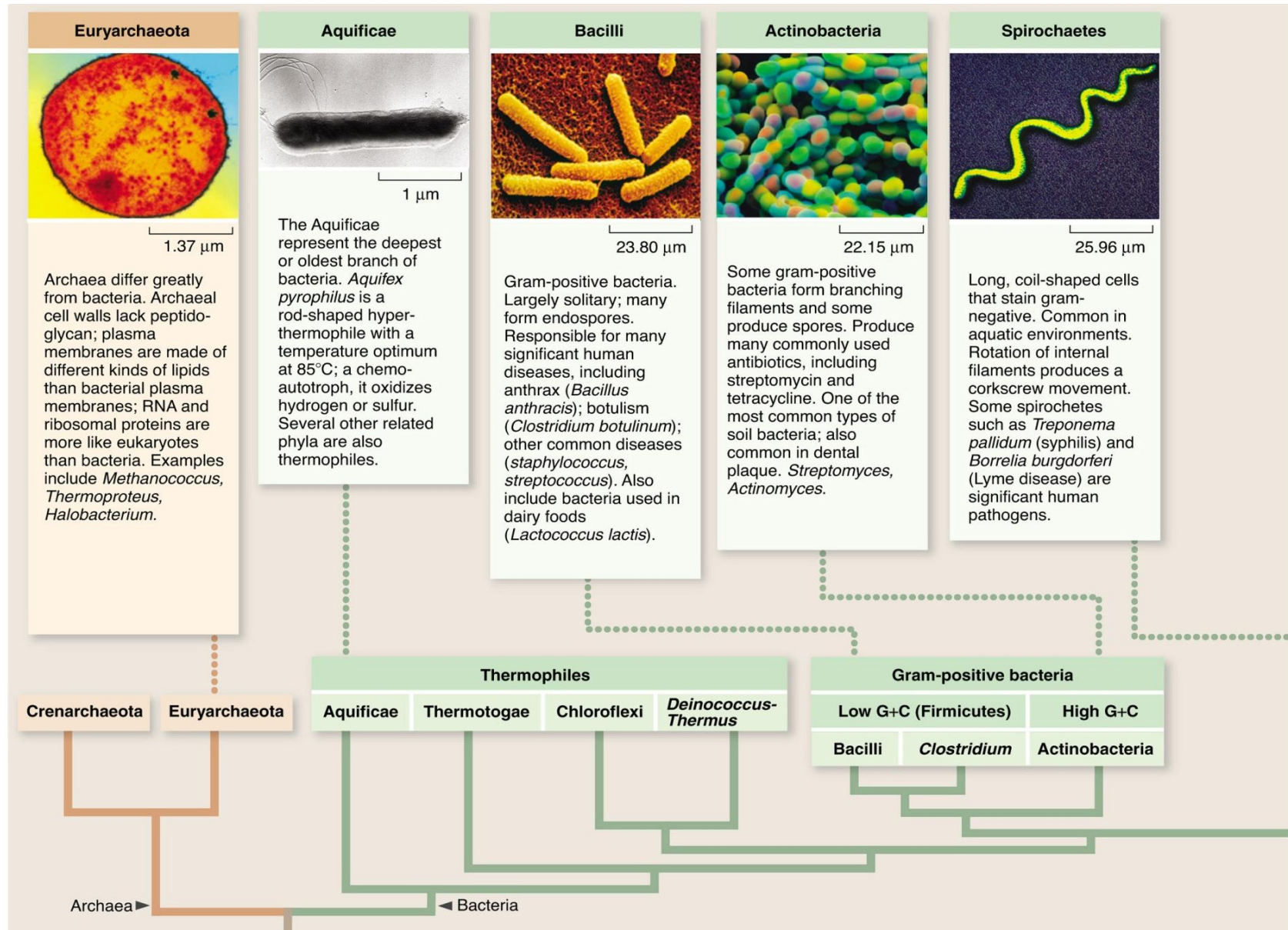
- Bergey's Manual of Systematic Bacteriology contains descriptions of all prokaryotic species currently identified
- Bergey's Manual Edition 2 (2001 on): improved, phylogenetic grouping; evolutionary relationships based on nucleic acid sequence
- There are now two major groups of Gram positive bacteria:
 - **Low G + C (<50%)**
 - **High G + C (>50%)**

Bases of DNA are: G = guanine(鸟嘌呤), C = cytosine(胞嘧啶), A = adenine(腺嘌呤), T = thymine(胸腺嘧啶)

$$\text{Mol\% G+C} = \frac{\text{G+C}}{\text{G+C+A+T}} \times 100$$

%G+C reflects DNA sequence, is easy to determine, is constant within a species and varies little within a genus; useful taxonomic characteristic of bacteria

Some major groups of Bacteria and Archaea (Prescott Fig 19.14)



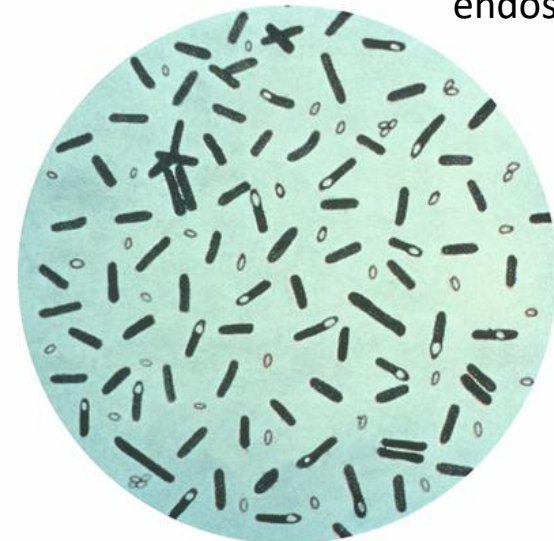
Phylum *Firmicutes* - Low G + C Gram-Positive Bacteria

1. Genus *Clostridium* (梭菌属)

- Gram-positive, rod-shaped
- **Obligate anaerobe**
 - does not tolerate O₂, dies in its presence
- Forms **endospores**
 - dormant, resistant to heat, chemicals etc.
- Fermentative
 - ferments sugars, amino acids, to gain energy
 - plays major role in decomposition and nitrogen cycling in anoxic marine sediments
- Species include serious human pathogens
 - *C. botulinum* causes botulism, a food-borne disease, can be associated with seafood



Subterminal
endospores



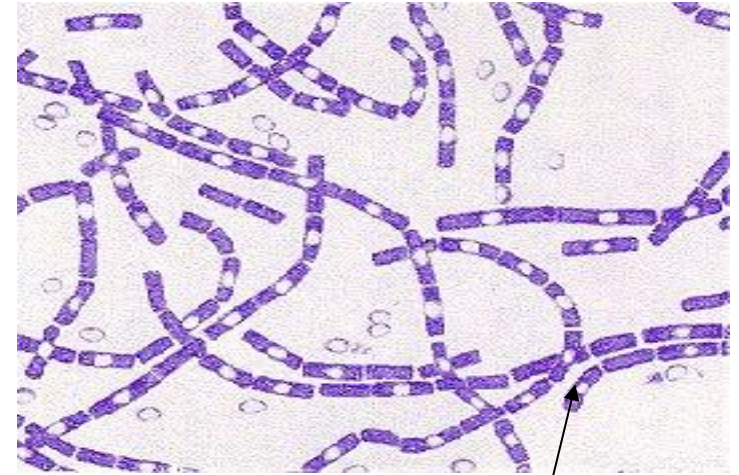
Phylum *Firmicutes* - Low G + C Gram-Positive Bacteria

2. Genus *Bacillus* (芽孢杆菌)

- Gram-positive, rod-shaped
- **Aerobic or facultative anaerobe**
 - can grow in presence or absence of O₂
- Forms **endospores**
- Some species produce antibiotics
- Found in soil, Isolated from marine sediments

Bacillus marinus (海洋芽孢杆菌)

Bacillus subtilis (枯草芽孢杆菌) used in the lab, best studied Gram positive



central
endospore

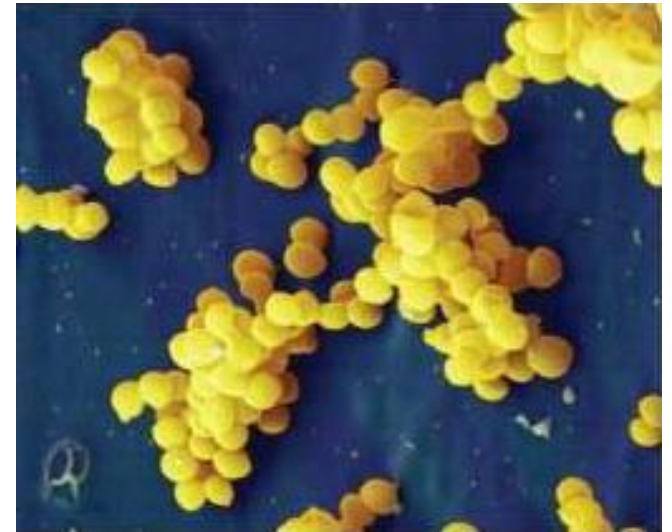
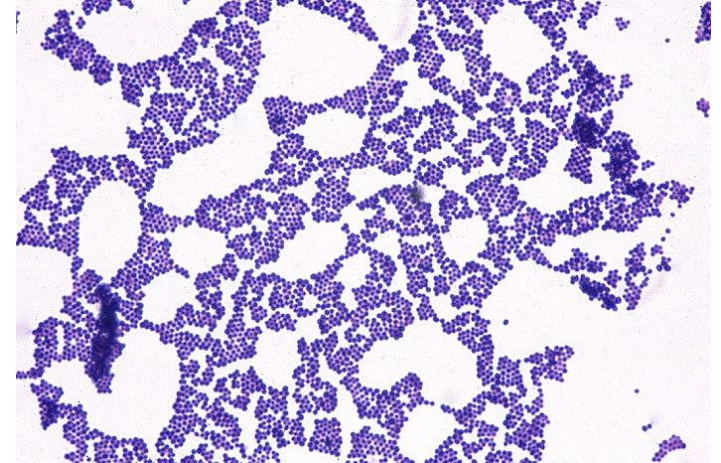
Phylum *Firmicutes* - Low G + C Gram-Positive Bacteria

3. Genus *Staphylococcus* (葡萄球菌)

- Gram-positive cocci
- Irregular clusters “bunch of grapes”
- Facultative anaerobes
- Associated with skin, mucous (粘液性的) membranes of vertebrates (脊椎动物)
- Minor in marine environments, but can contaminate fish during processing

Staphylococcus epidermidis (表皮葡萄球菌)
–used in our labs, on our fingers! Only causes infections in patients with lowered immunity.

Staphylococcus aureus (金黄色葡萄球菌) –
human pathogen



Phylum *Firmicutes* - Low G + C Gram-Positive Bacteria

4. Genus *Streptococcus* (链球菌)

- Gram-positive cocci
- Cells in pairs or chains
- Facultative anaerobes

Streptococcus iniae: cause of disease in marine warm-water fish culture, and infections in humans working in aquaculture



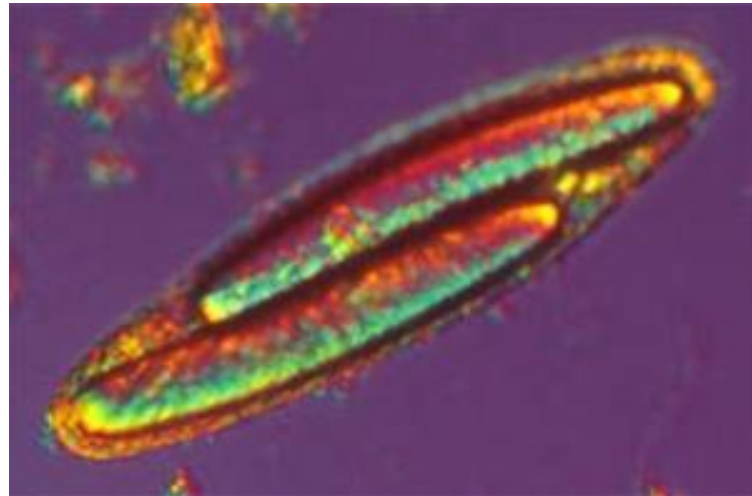
S. iniae



Phylum *Firmicutes* - Low G + C Gram-Positive Bacteria

5. Genus *Epulopiscium*

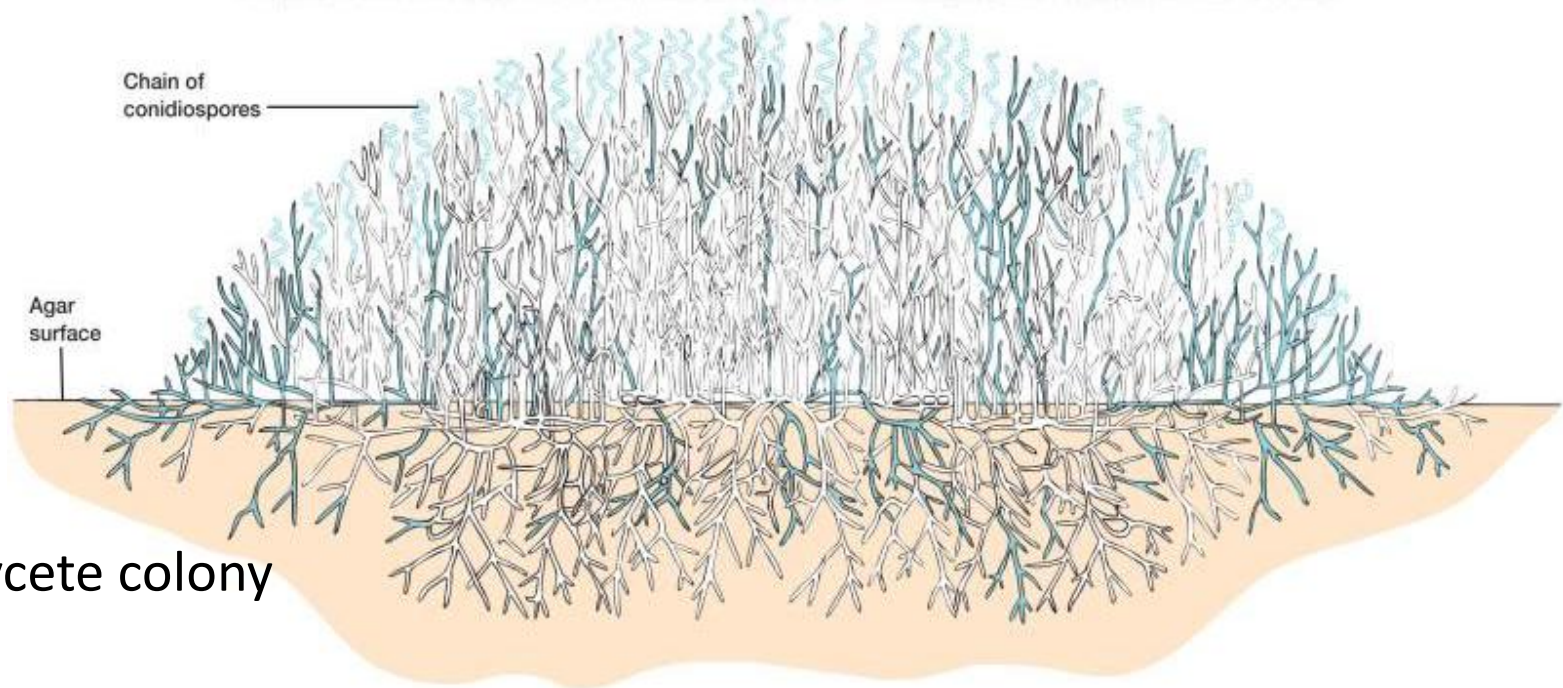
- One of the largest bacteria known – up to 600 μm
- *Epulopiscium fishelsoni* (费氏刺骨鱼菌): found in large numbers in the intestinal tracts of surgeonfish (刺尾鱼) on Great Barrier Reef大堡礁 (N. Australia) and Red Sea
- Unusual life cycle
 - Viviparous (胎生的) = new cells are formed inside the parent cell, which lyses to release them



Phylum *Actinobacteria* - High G + C Gram-Positive Bacteria

Actinomycetes 放线菌 (a general name: many members of the phylum are Actinomycetes)

- Gram-positive, aerobic, form branching filaments or hyphae and asexual spores (not endospores)
- Many resemble (像) fungi in morphology (形态学)
- Primarily soil inhabitants, but widely distributed in marine sediments
- Produce most of the medically useful antibiotics

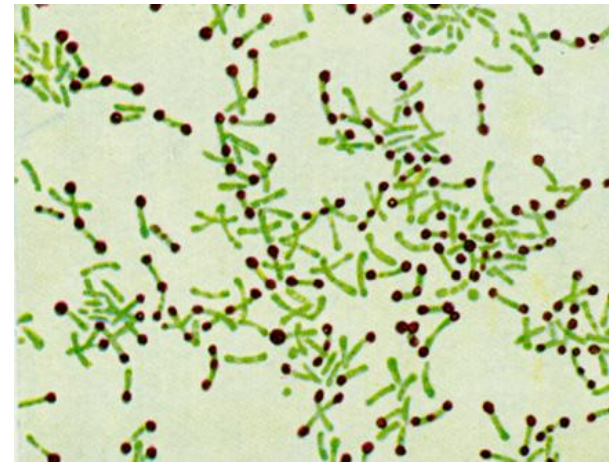
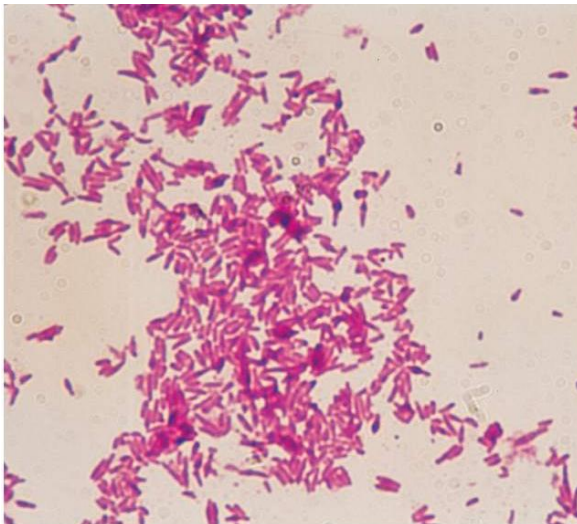
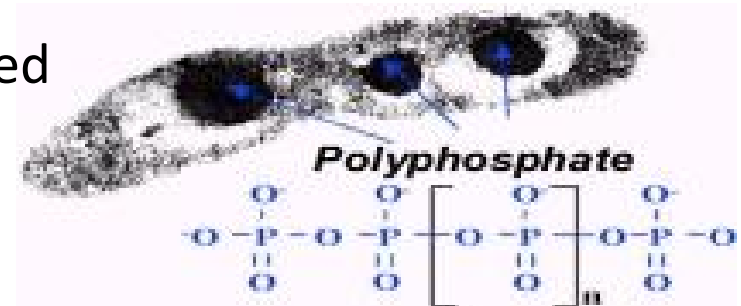


An actinomycete colony

Phylum *Actinobacteria* - High G + C Gram-Positive Bacteria

1. Genus *Corynebacterium* (棒状杆菌)

- Gram-positive straight, curved or club-shaped rods
- Angular arrangements of cells
 - “chinese letters”
- Form metachromatic 异染性的(volutin) granules
 - energy and phosphate reserve



Granules stained brown

Phylum *Actinobacteria* - High G + C Gram-Positive Bacteria

2. Genus *Mycobacterium* (分支杆菌属)

- Gram-positive rods - sometimes branched/filaments
- Grow very slowly - up to 40 days → colony
- Waxy (蜡状) cell wall has high lipid content
- Widely distributed on surfaces such as sediments, coral, fish and algae



M. tuberculosis colonies

M. tuberculosis (结核分枝杆菌): Tuberculosis (TB) - global problem - 20% infected, 3 million deaths per year

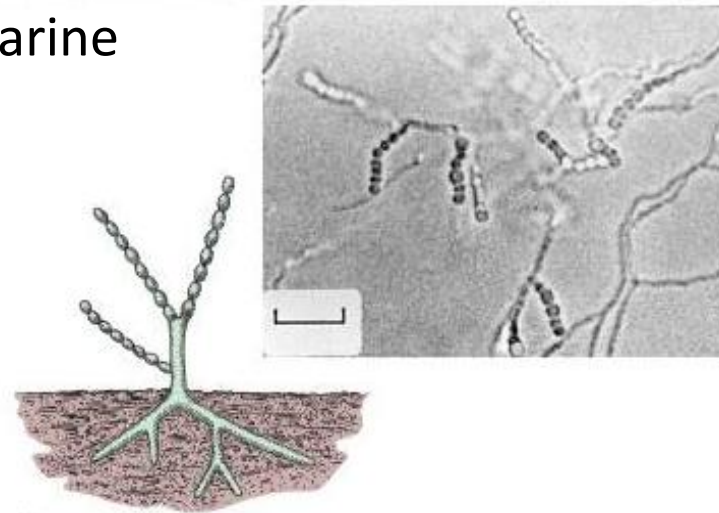
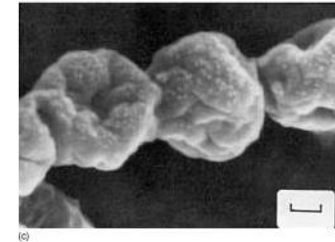
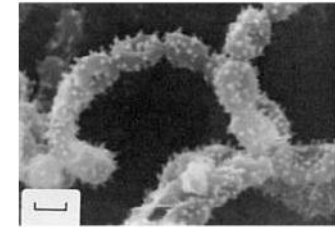
M. marinum (海洋分枝杆菌): pathogen of fish and marine mammals which can be transmitted to humans

Phylum *Actinobacteria* - High G + C Gram-Positive Bacteria

3. Genus *Streptomyces* (链霉菌属)

- Gram-positive rods
- Extensively branched filaments (mycelium)
- Aerial hyphae (气生菌丝) with chains of spores (not endospores)
- Found in soil, marine sediments
- Synthesise numerous **antibiotics**
- Useful bioactive compounds from marine *Streptomyces* are being investigated

Typical
streptomycete
morphology and
spore variety



Summary

1. Gram-negative bacteria are classified into Non-Proteobacteria and Proteobacteria
2. Gram-positive bacteria are classified into low G + C and high G + C
3. *Cyanobacteria* carry out oxygenic photosynthesis; purple and green bacteria use anoxygenic photosynthesis; all cause ocean blooms
4. Proteobacteria are divided into 5 subgroups (α , β , γ , δ , ϵ) with numerous important marine genera
5. Magnetotactic bacteria swim towards microaerophilic sediments
6. The low G + C *Firmicutes* include endospore-forming genera *Bacillus* and *Clostridium*, as well as *Staphylococcus*, *Streptococcus* and the large marine genus *Epulopiscium*
7. The high G + C bacteria include the Actinomycete genera *Corynebacterium*, *Mycobacterium* and *Streptomyces*